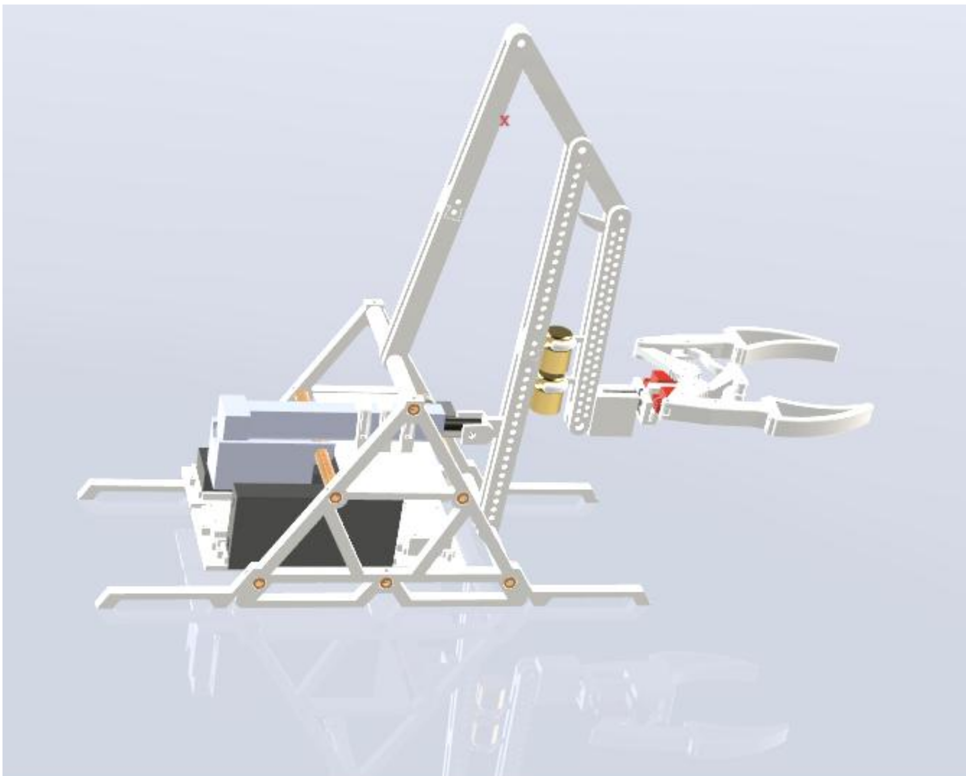
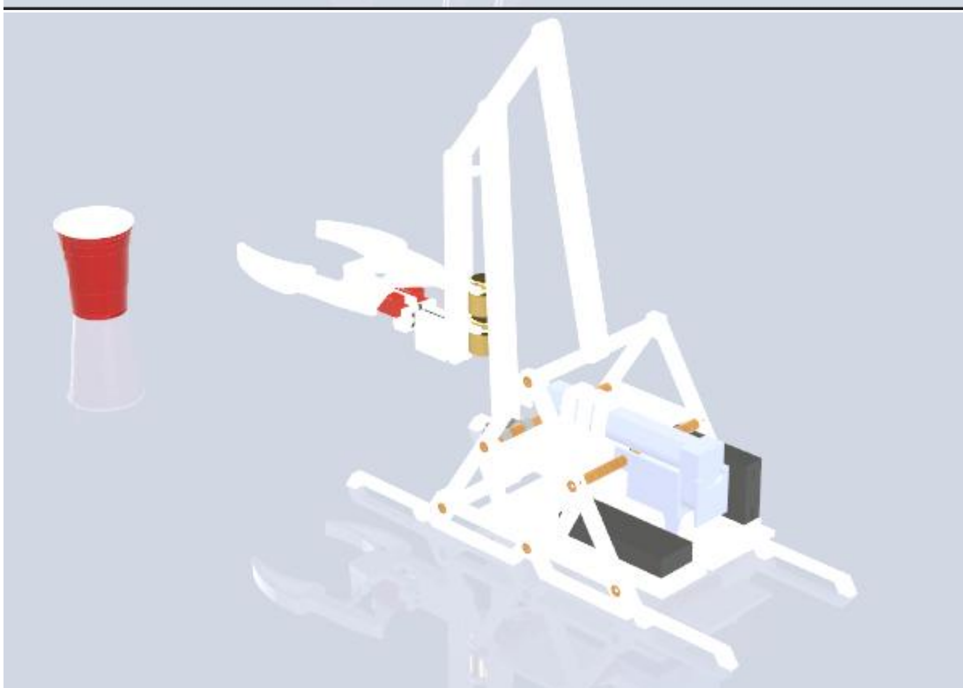
1. OVERVIEW

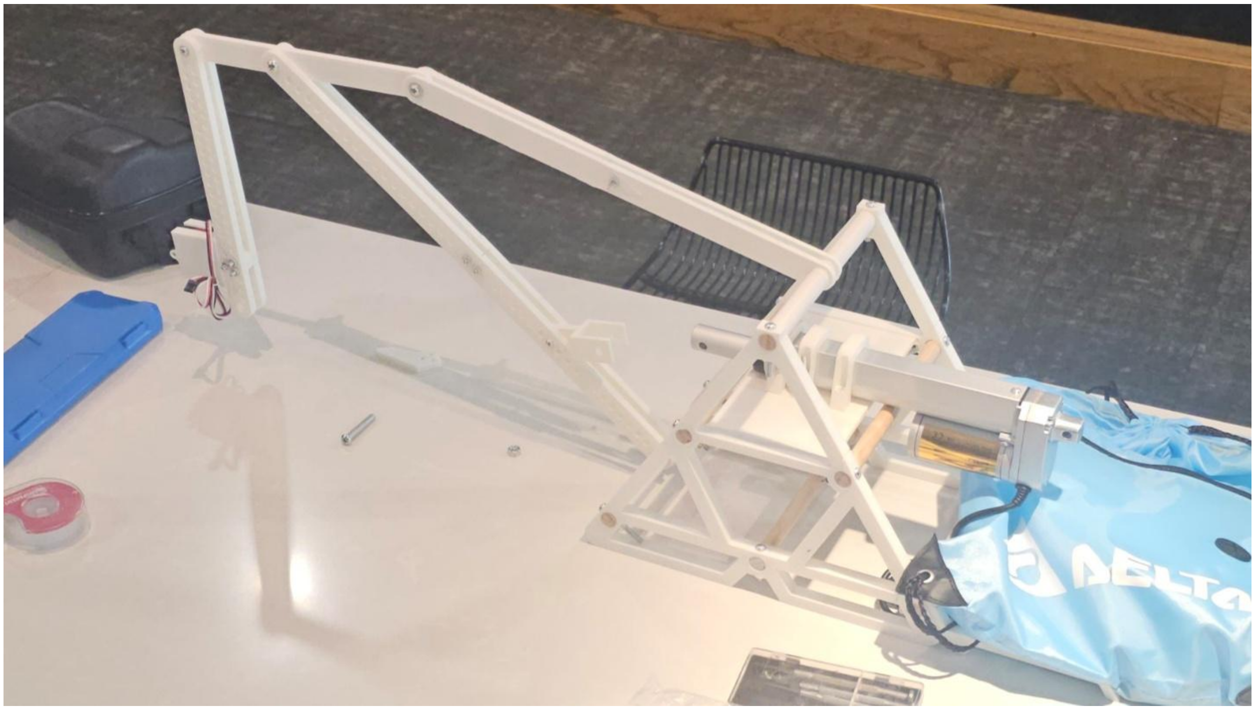
1.1. Device images

SolidWorks Visualize:



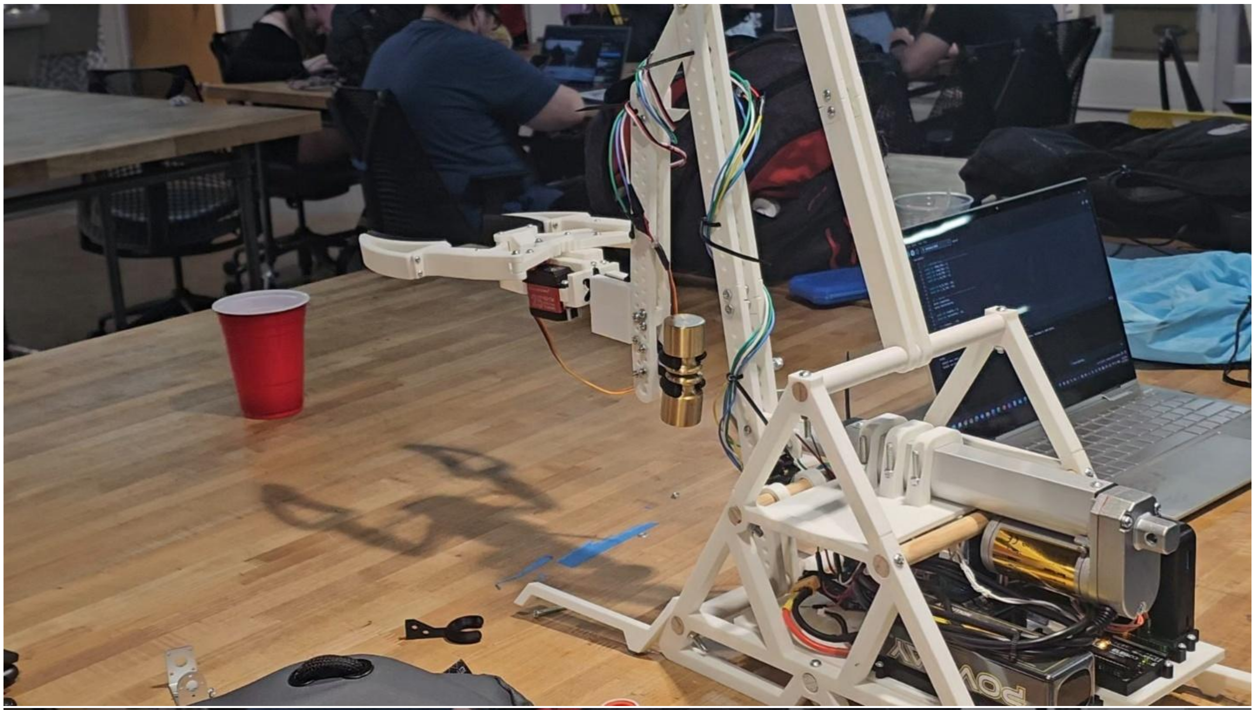


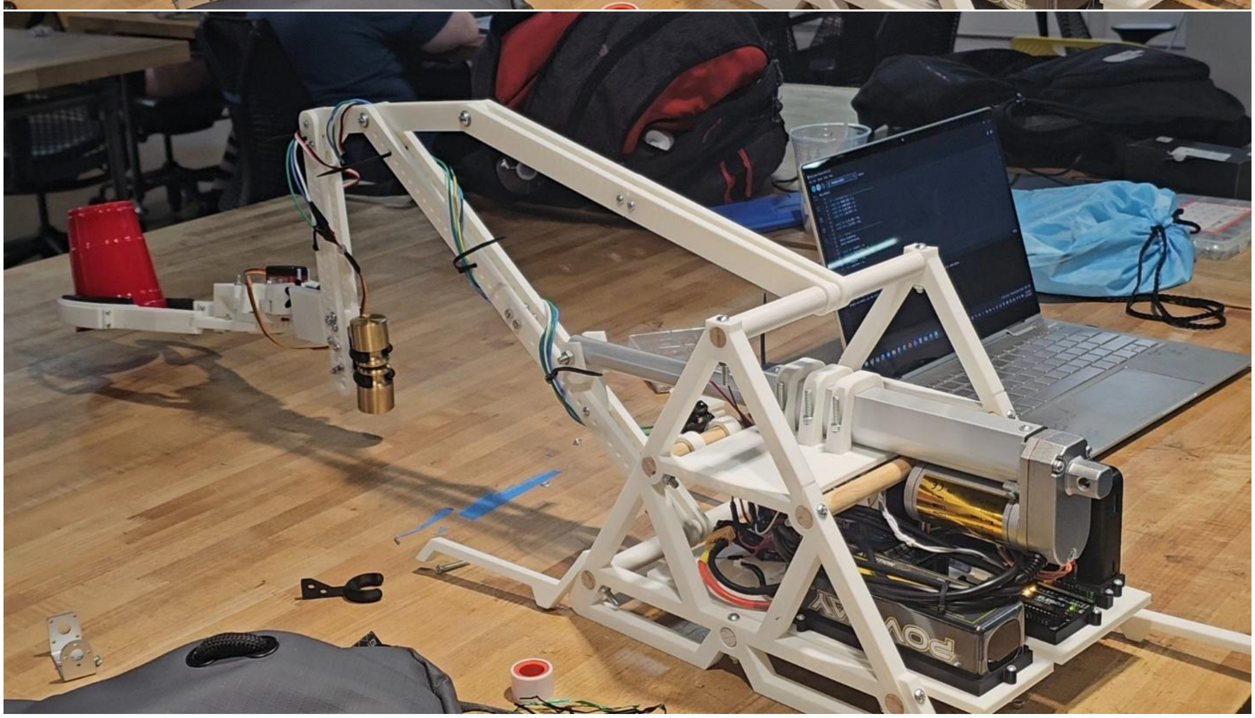
(a) Overall View of Prototype:



(b) One Part Being Fabricated

(c) Device in Operation.





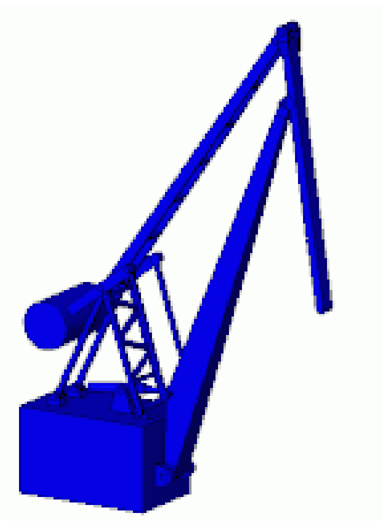
1.2. Design references

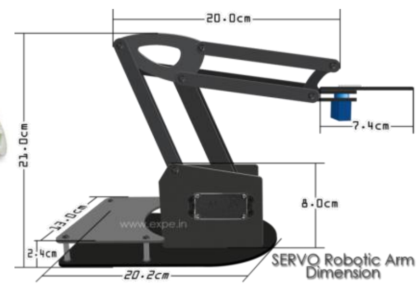
One major modification we made to the 'level-luffing' crane design, is the ability for our front arm to dangle as it extends forward. The reason we did this is because we needed to extend the reaching arm to make it lower to the ground, but this caused our claw to clip into the ground. So we allowed the arm to pivot, to keep the claw level

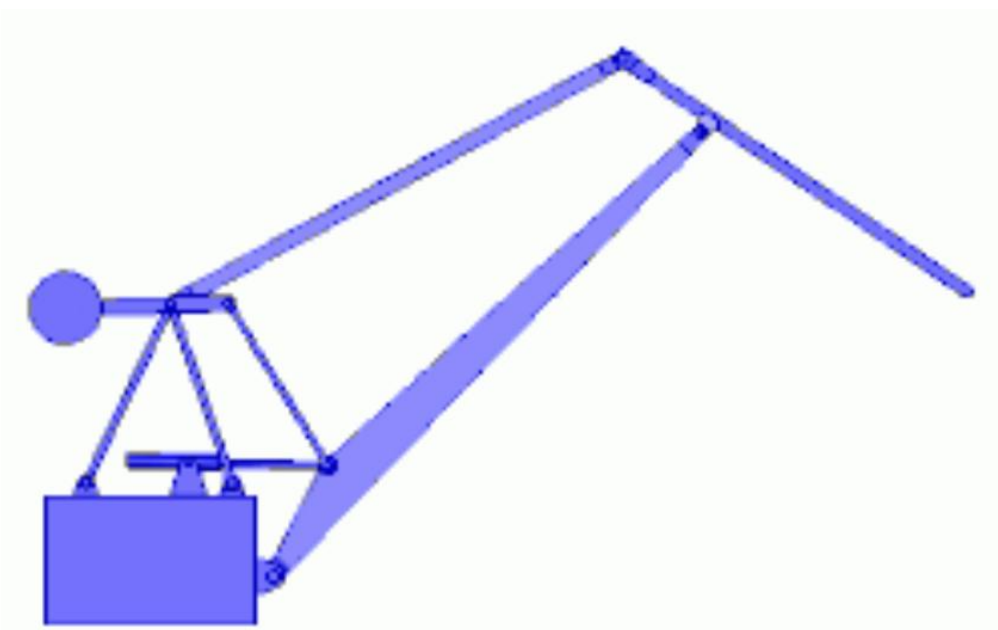
We also removed the counter-weight and added legs for stability.

We also modified the base of the crane into a more open, and more cost-effective frame.

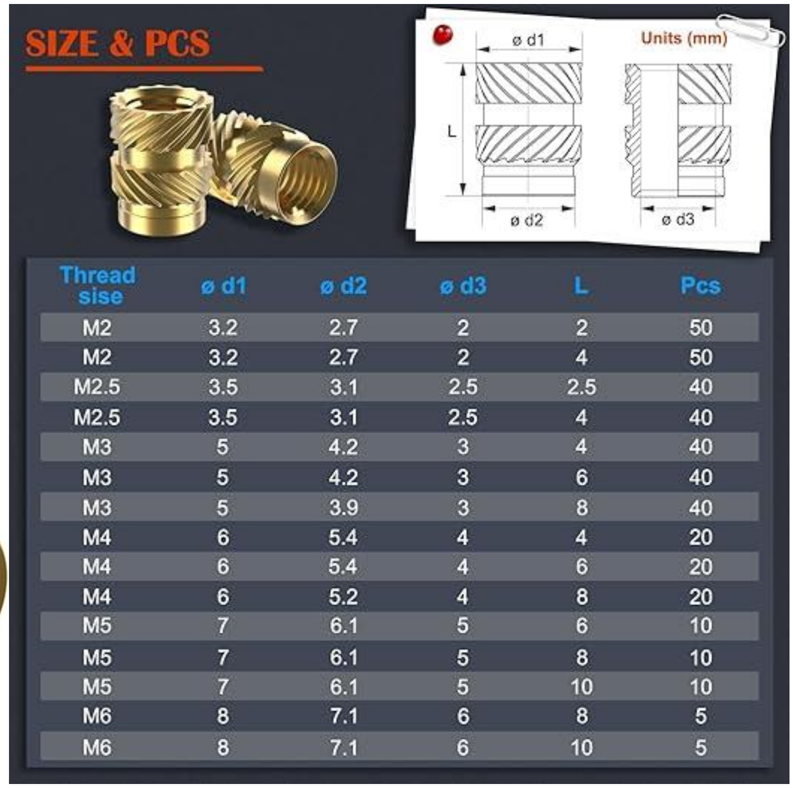
https://www.youtube.com/watch?v=goaKXiSi53g







1.3. COTS component



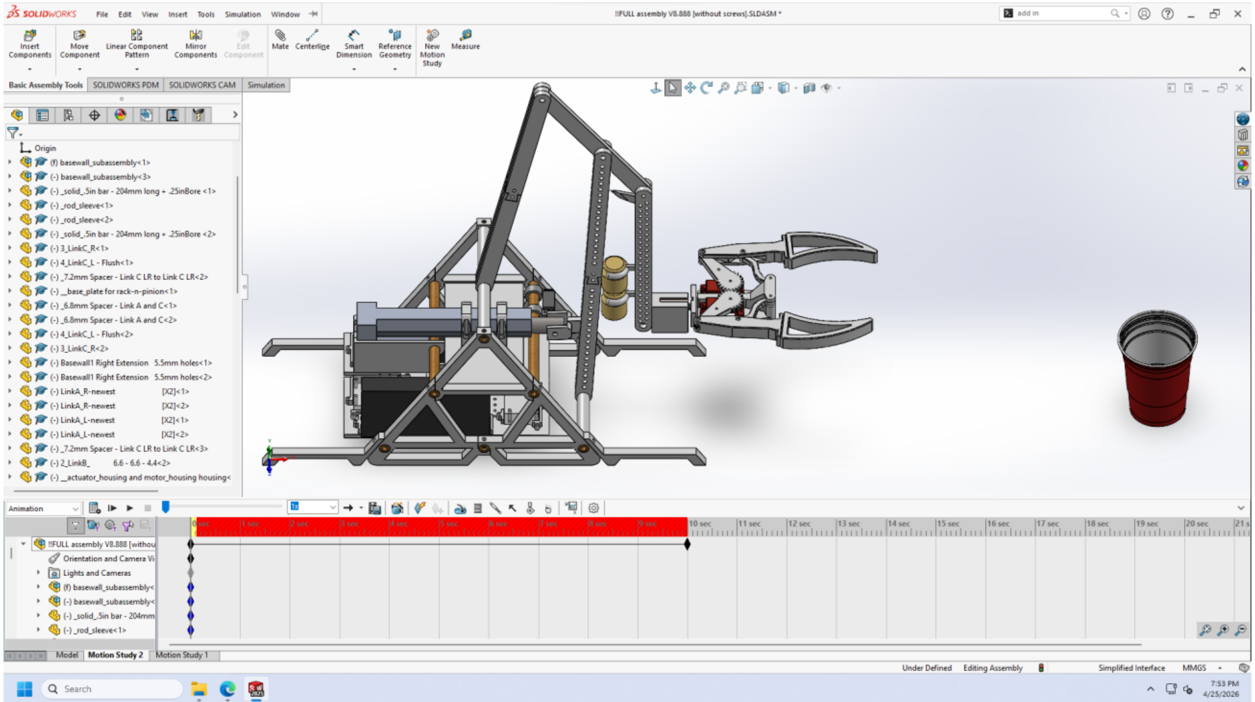
Thread sise	ø d1	ø d2	ø d3	L	Pcs
M2	3.2	2.7	2	2	50
M2	3.2	2.7	2	4	50
M2.5	3.5	3.1	2.5	2.5	40
M2.5	3.5	3.1	2.5	4	40
M3	5	4.2	3	4	40
M3	5	4.2	3	6	40
M3	5	3.9	3	8	40
M4	6	5.4	4	4	20
M4	6	5.4	4	6	20
M4	6	5.2	4	8	20
M5	7	6.1	5	6	10
M5	7	6.1	5	8	10
M5	7	6.1	5	10	10
M6	8	7.1	6	8	5
M6	8	7.1	6	10	5

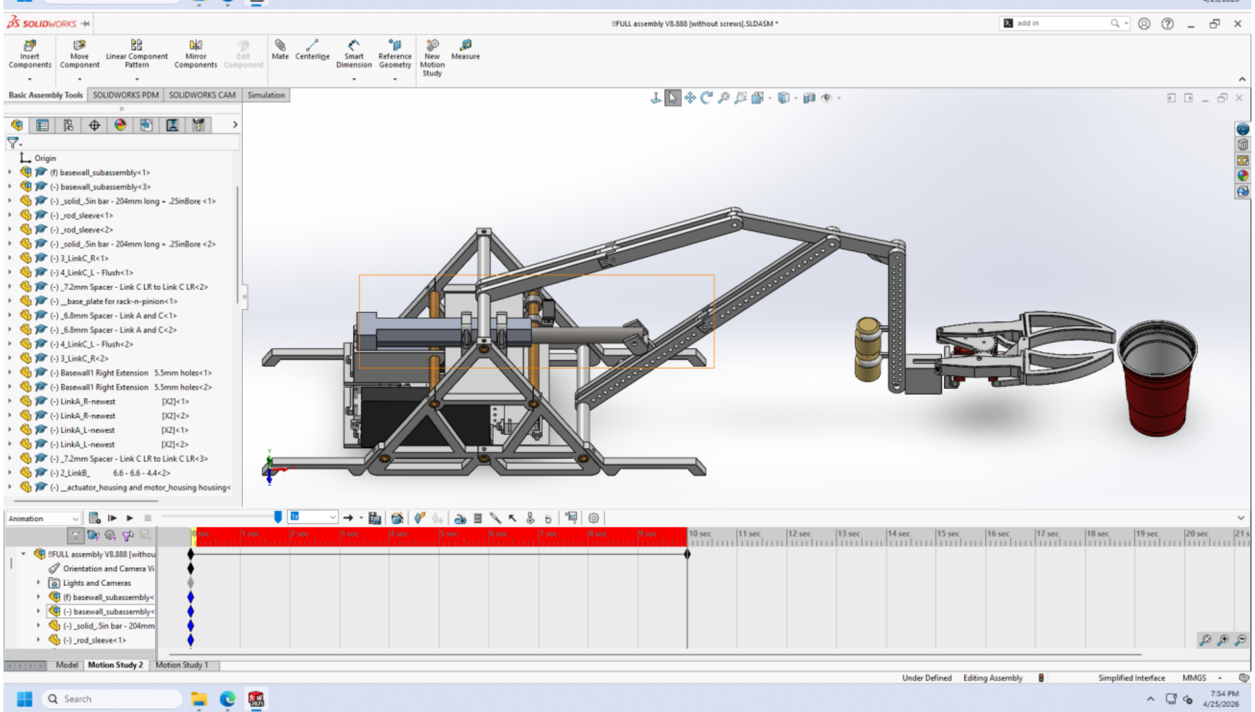
Used: 4x 6mm long M4 inserts the in claw's grabber's

Item: 400Pcs Threaded Inserts, M2 M2.5 M3 M4 M5 M6 Assortment Kit
Cost: $15.98
Manufacturer: Ktehloy   [No Official Website]
Distributor: Amazon
Purchase Link: https://a.co/d/04GWM11r

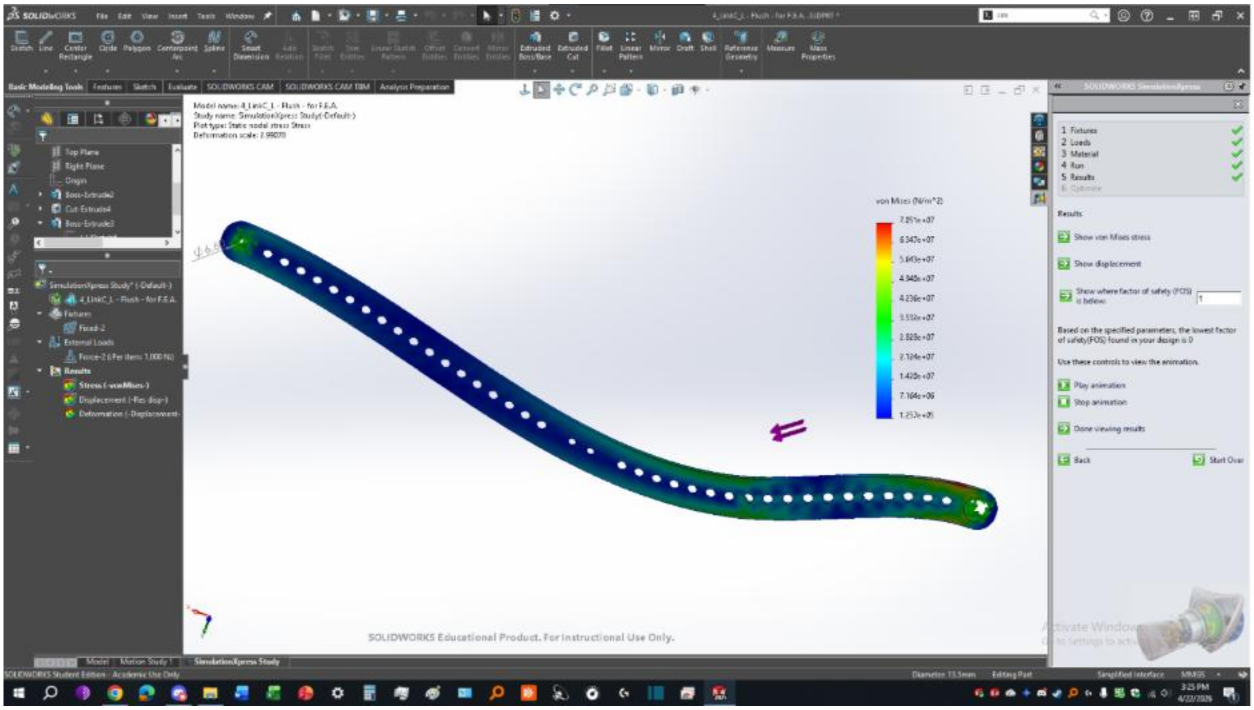
2. SIMULATION

2.1. Animation





2.2. Simulation Analysis



The part was loaded at the loaded at the 11th small hole from the bottom right, similar to how the linear actuator will apply a force to the link at that specific hole.

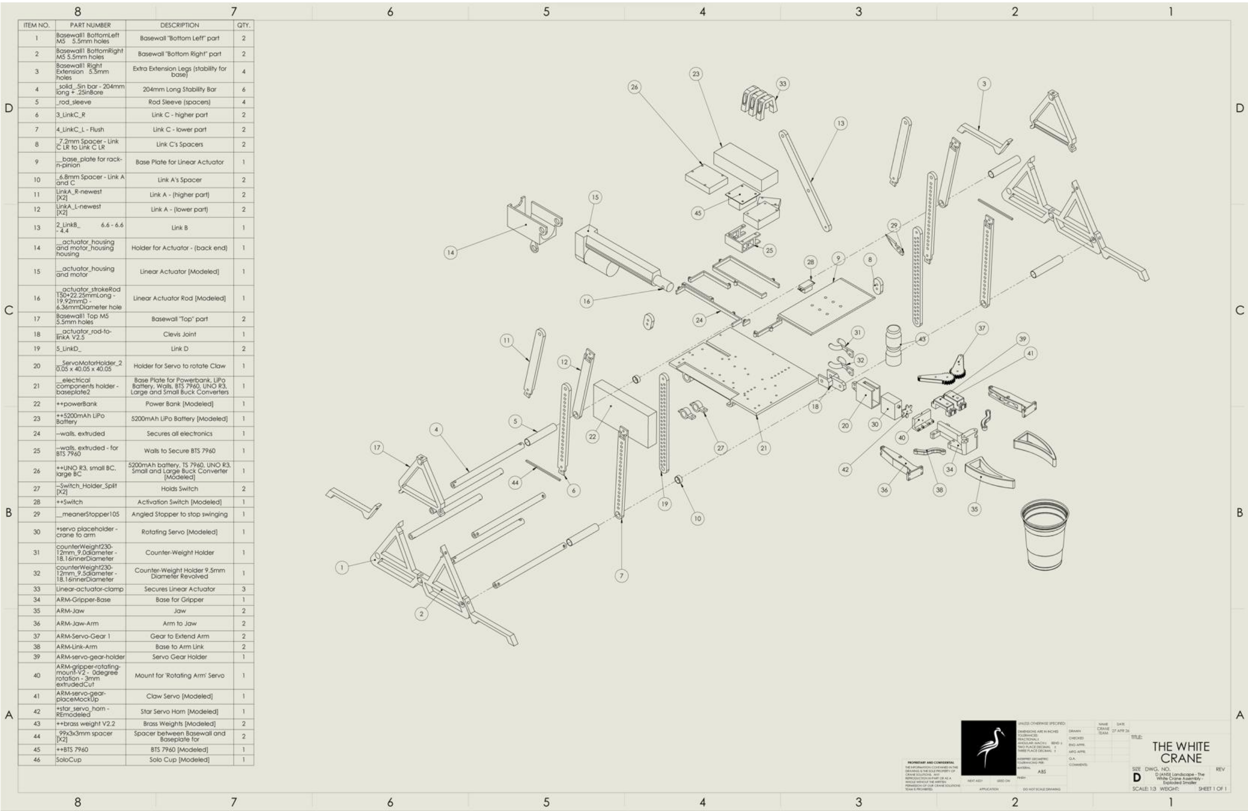
The part had one fixed point at the top left, which connects to a link, and the other fixed point at the bottom right was fixed to the base's rod.

The force we applied was 1000 Newtons, since the maximum force the actuator could produce was 1000 Newtons, and we wanted to see the force at that spot if the whole link is fixed.

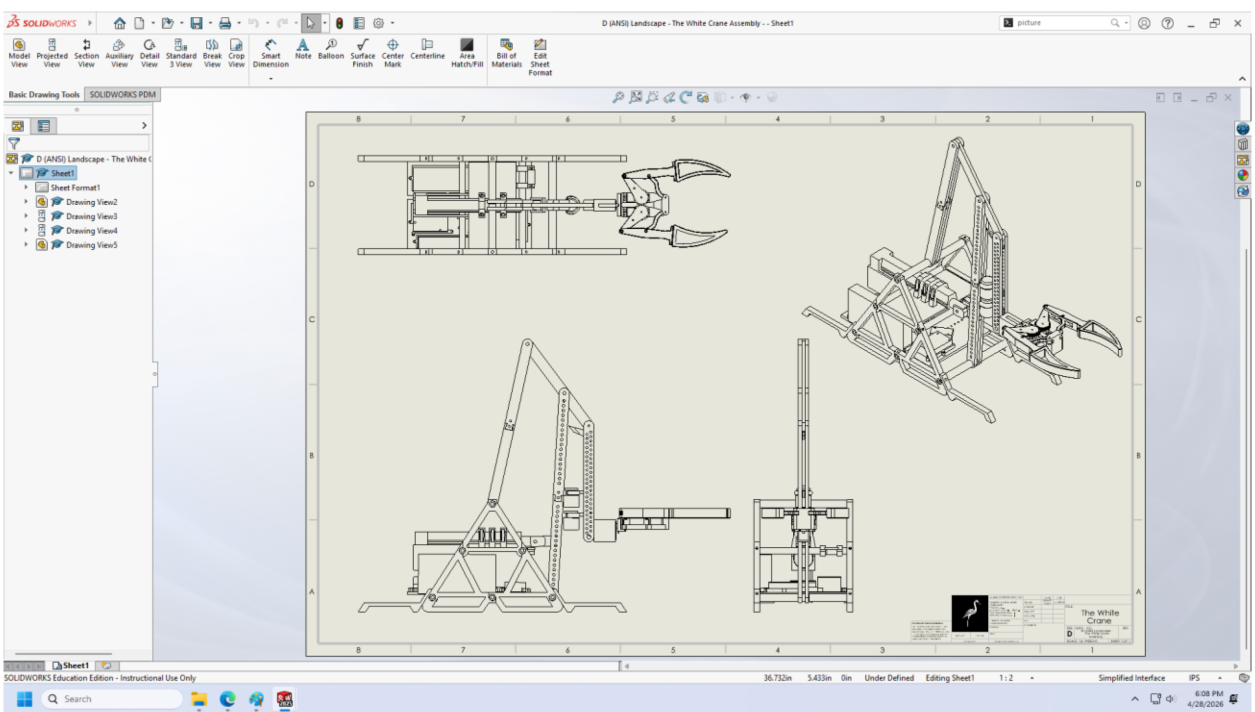
The part that's fixed has red color represents that it needs to be worked on as it experiences the most pressure that may cause it to deform and apart from that the FOI of that part could be changed to reduce the deformation rate. Overall, even though the part experiences a 1000N load it has only the fixed bottom area that experiences maximum load and could potentially deform which could be improved as well. However, the part in general has a lot of blue color that is the lowest stress area along with some greens that has the second highest deformation rate. The Von meses table on the right represents this data.

3. ENGINEERING WORKING DRAWINGS

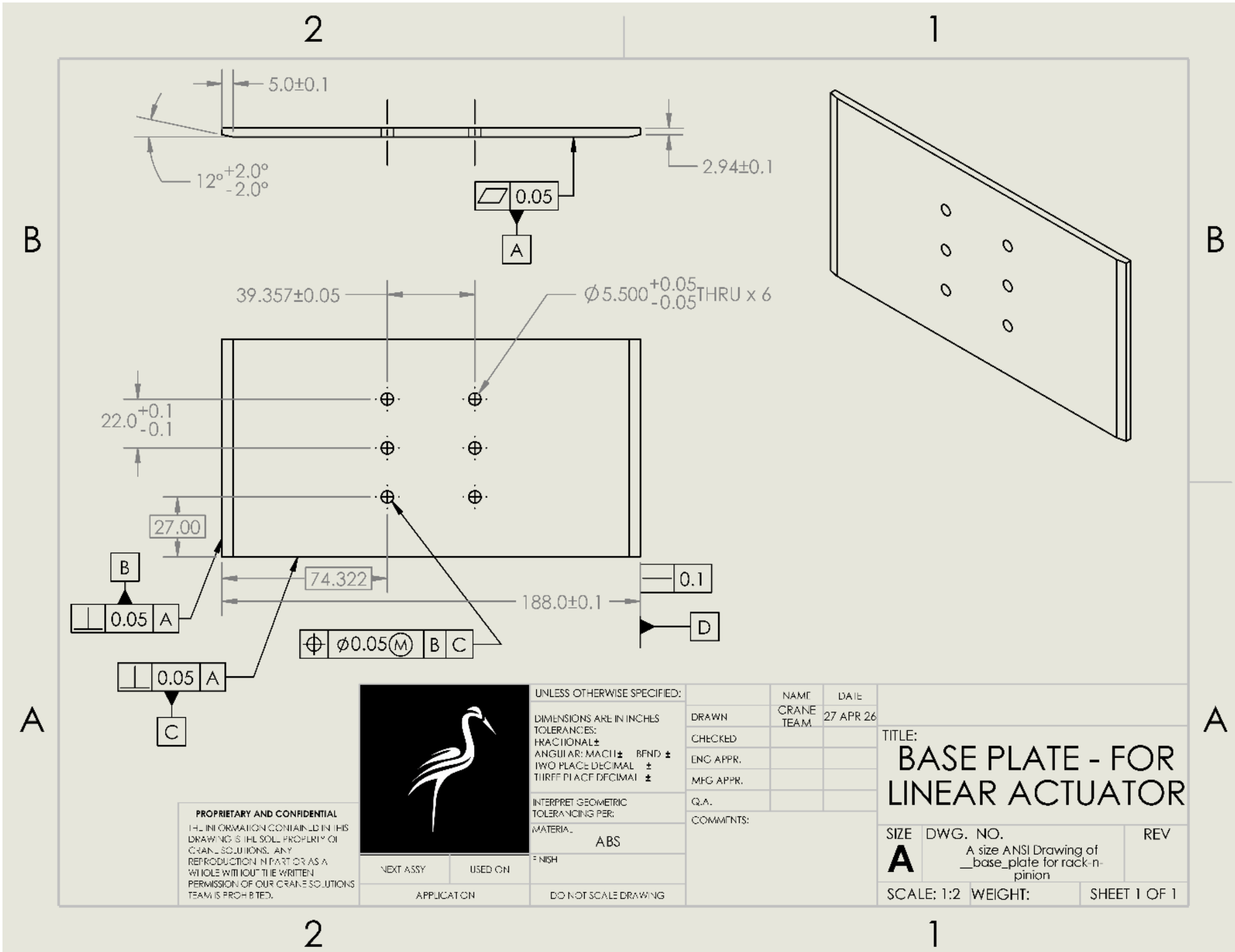
3.1. Assembly drawing



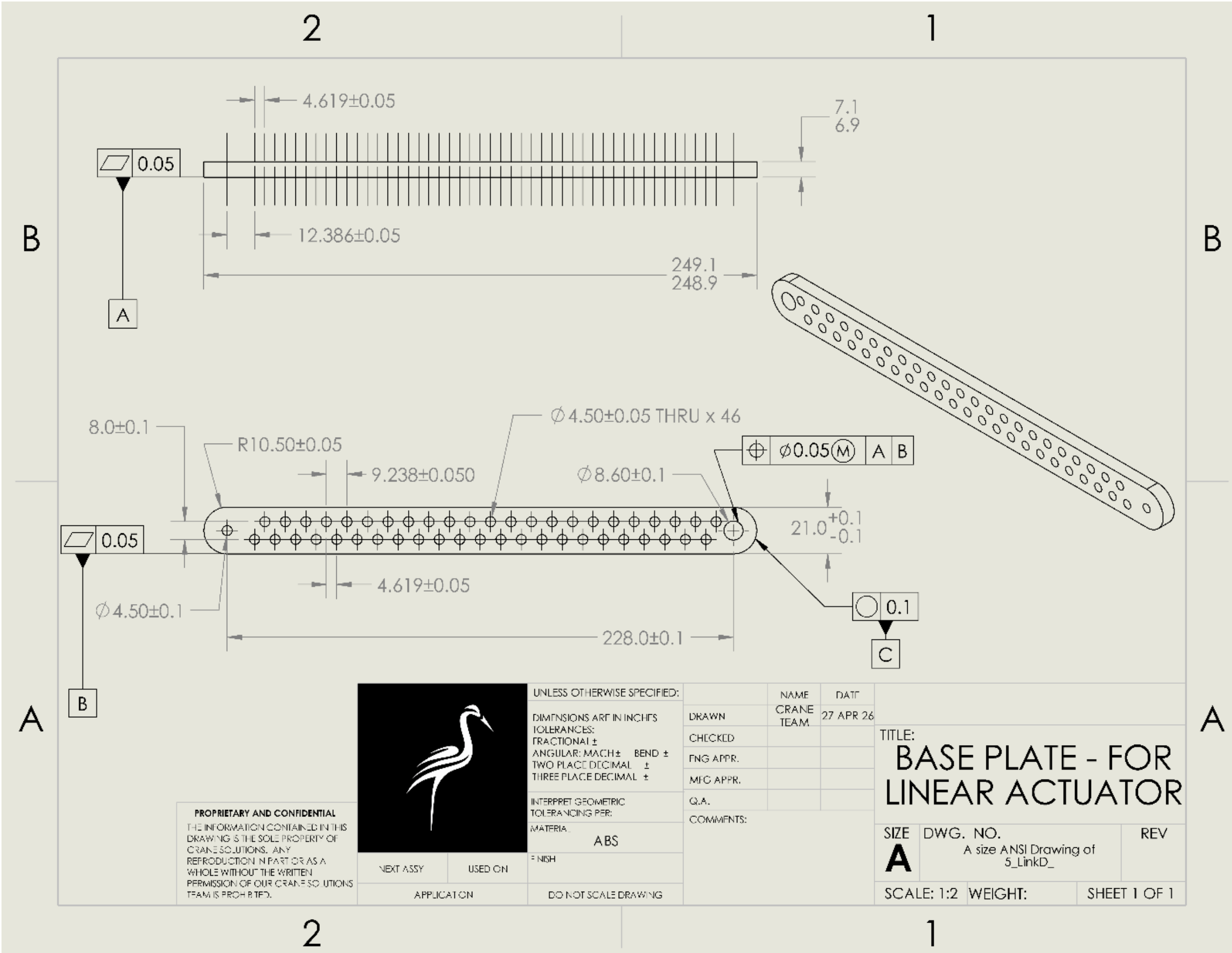
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Basewall1 BottomLeft M5 5.5mm holes	Basewall "Bottom Left" part	2
2	Basewall1 BottomRight M5 5.5mm holes	Basewall "Bottom Right" part	2
3	Basewall1 Right Extension 5.5mm holes	Extra Extension Legs (stability for base)	4
4	_solid_.5in bar - 204mm long + .25inBore	204mm Long Stability Bar	6
5	_rod_sleeve	Rod Sleeve (spacers)	4
6	3_LinkC_R	Link C - higher part	2
7	4_LinkC_L - Flush	Link C - lower part	2
8	_7.2mm Spacer - Link C LR to Link C LR	Link C's Spacers	2
9	__base_plate for rack-n-pinion	Base Plate for Linear Actuator	1
10	_6.8mm Spacer - Link A and C	Link A's Spacer	2
11	LinkA_R-newest [X2]	Link A - (higher part)	2
12	LinkA_L-newest [X2]	Link A - (lower part)	2
13	2_LinkB_ 6.6 - 6.6 - 4.4	Link B	1
14	__actuator_housing and motor_housing housing	Holder for Actuator - (back end)	1
15	__actuator_housing and motor	Linear Actuator [Modeled]	1
16	__actuator_strokeRod 150+22.25mmLong - 19.92mmD - 6.36mmDiameter hole	Linear Actuator Rod [Modeled]	1
17	Basewall1 Top M5 5.5mm holes	Basewall "Top" part	2
18	__actuator_rod-to-linkA V2.5	Clevis Joint	1
19	5_LinkD_	Link D	2
20	__ServoMotorHolder_20.05 x 40.05 x 40.05	Holder for Servo to rotate Claw	1
21	__electrical components holder - baseplate2	Base Plate for Powerbank, LiPo Battery, Walls, BTS 7960, UNO R3, Large and Small Buck Converters	1
22	++powerBank	Power Bank [Modeled]	1
23	++5200mAh LiPo Battery	5200mAh LiPo Battery [Modeled]	1
24	--walls, extruded	Secures all electronics	1
25	--walls, extruded - for BTS 7960	Walls to Secure BTS 7960	1
26	++UNO R3, small BC, large BC	5200mAh battery, TS 7960, UNO R3, Small and Large Buck Converter [Modeled]	1
27	--Switch_Holder_Split [X2]	Holds Switch	2
28	++Switch	Activation Switch [Modeled]	1
29	__meanerStopper105	Angled Stopper to stop swinging	1
30	+servo placeholder - crane to arm	Rotating Servo [Modeled]	1
31	counterWeight230-12mm_9.0diameter - 18.16innerDiameter	Counter-Weight Holder	1
32	counterWeight230-12mm_9.5diameter - 18.16innerDiameter	Counter-Weight Holder 9.5mm Diameter Revolved	1
33	Linear-actuator-clamp	Secures Linear Actuator	3
34	ARM-Gripper-Base	Base for Gripper	1
35	ARM-Jaw	Jaw	2
36	ARM-Jaw-Arm	Arm to Jaw	2
37	ARM-Servo-Gear 1	Gear to Extend Arm	2
38	ARM-Link-Arm	Base to Arm Link	2
39	ARM-servo-gear-holder	Servo Gear Holder	1
40	ARM-gripper-rotating-mount-V2 - 0degree rotation - 3mm extrudedCut	Mount for 'Rotating Arm' Servo	1
41	ARM-servo-gear-placeMockUp	Claw Servo [Modeled]	1
42	+star_servo_horn - REmodeled	Star Servo Horn [Modeled]	1
43	++brass weight V2.2	Brass Weights [Modeled]	2
44	_99x3x3mm spacer [X2]	Spacer between Basewall and Baseplate for	2
45	++BTS 7960	BTS 7960 [Modeled]	1
46	SoloCup	Solo Cup [Modeled]	1



3.2. Detail part drawing package #1



3.3. Detail part drawing package #2



Zip through MS OneDrive: *[The White Crane - Mechanism + .zip]* (only visible to people within UTA)

Zip through Google Drive:
https://drive.google.com/file/d/1ZHzc2OXGPN2vK3tYixMO6ENHRjNZa7WJ/view?usp=sharing